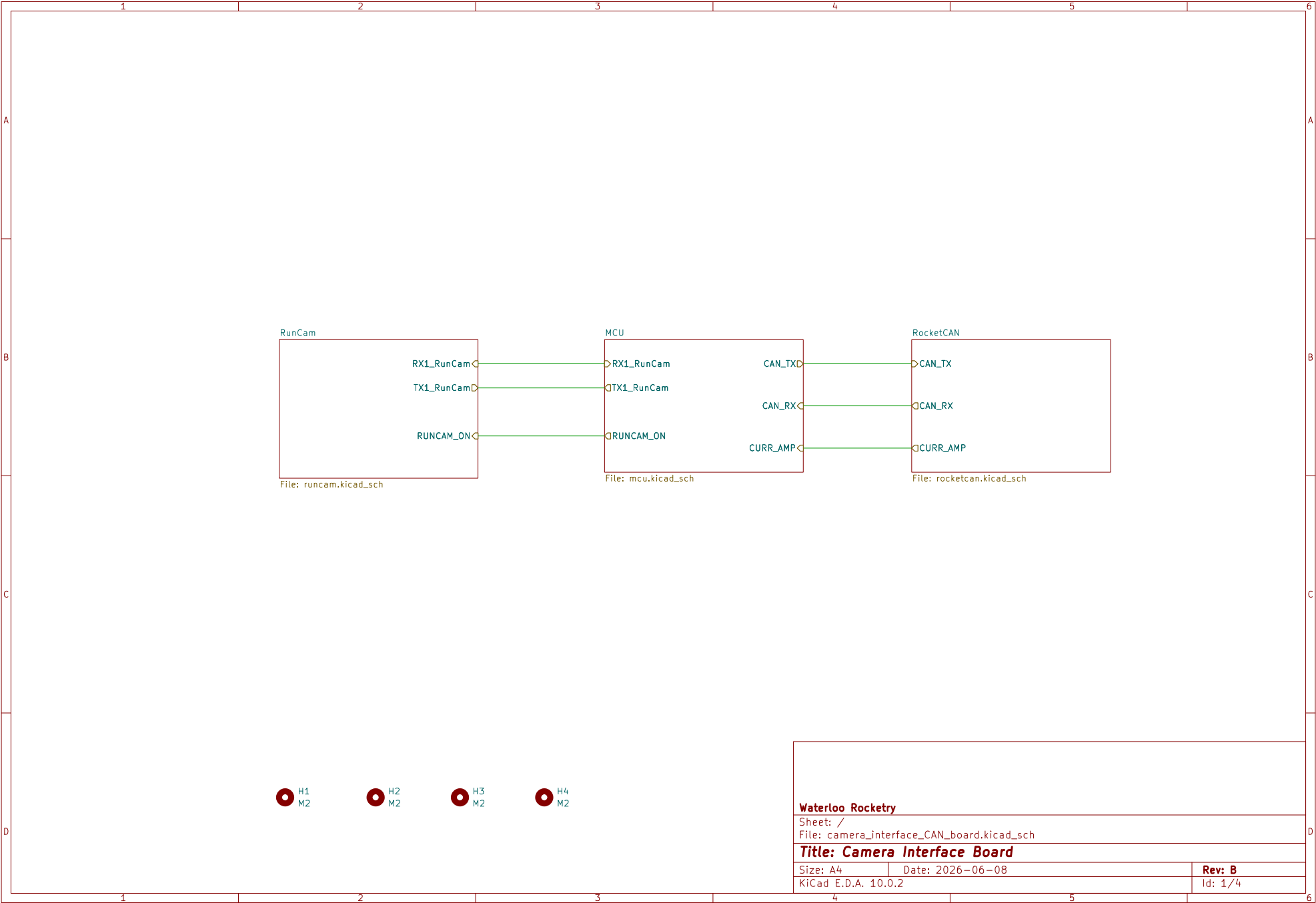
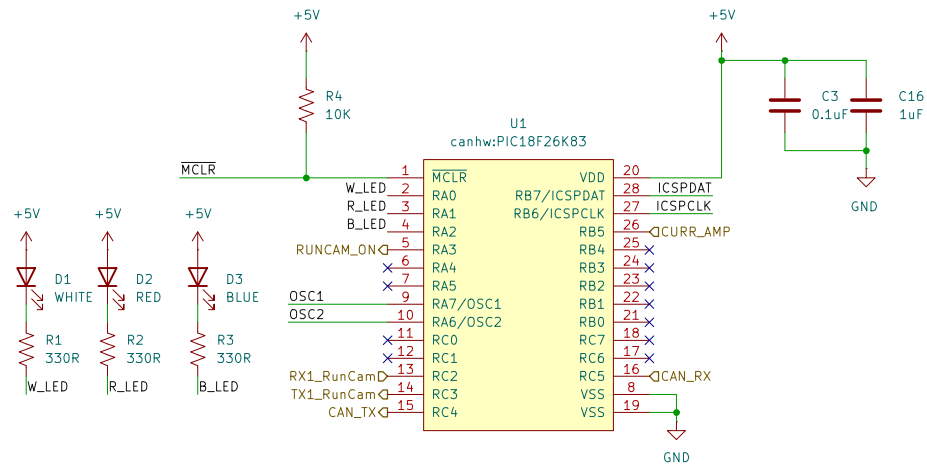
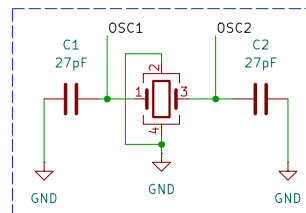
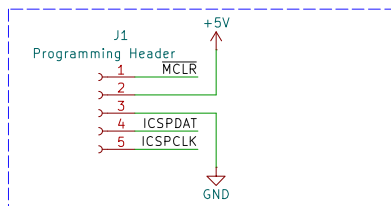






CL = 18pF from data sheet, Cstray = 5pF
 $CL = (C1 \times C2) / (C1 + C2) + Cstray$
 $C1 = C2$
 $18 = C/2 + 5$
 $C = 26pF$
 $C = 22pF$ from E12
 12 MHz crystal oscillator



Y1
 ABM8G-12.000MHZ-18-D2Y-T

Waterloo Rocketry

Sheet: /MCU/
 File: mcu.kicad_sch

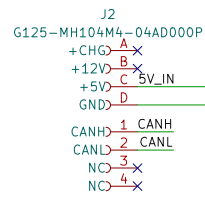
Title: Camera Interface Board

Size: A4 Date: 2026-06-08

KiCad E.D.A. 10.0.2

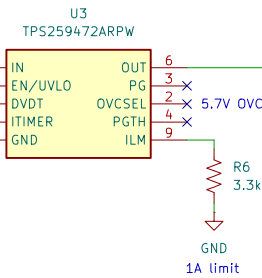
Rev: B

Id: 2/4

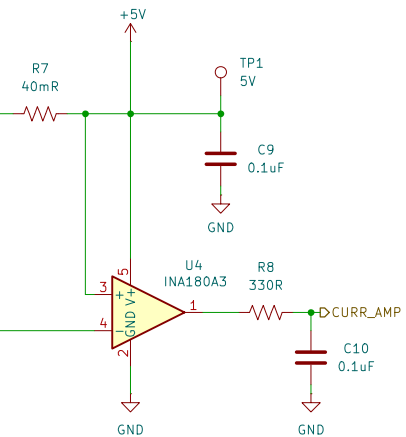


13Vrwm to survive accidental 12V input

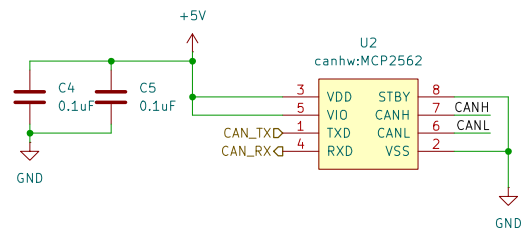
Datasheet: for supply voltages greater than 5V or systems which can be exposed to reverse polarity on input supply, it is recommended to use a pull-up resistor with a minimum value of 350kΩ.



Auto-retry version. Latch-off also available



$1A \cdot R_{sense} \cdot 100V/V = 4.096V$
 $R=0.041R$, choose $R=0.04$



Waterloo Rocketry

Sheet: /RocketCAN/
 File: rocketcan.kicad_sch

Title: Camera Interface Board

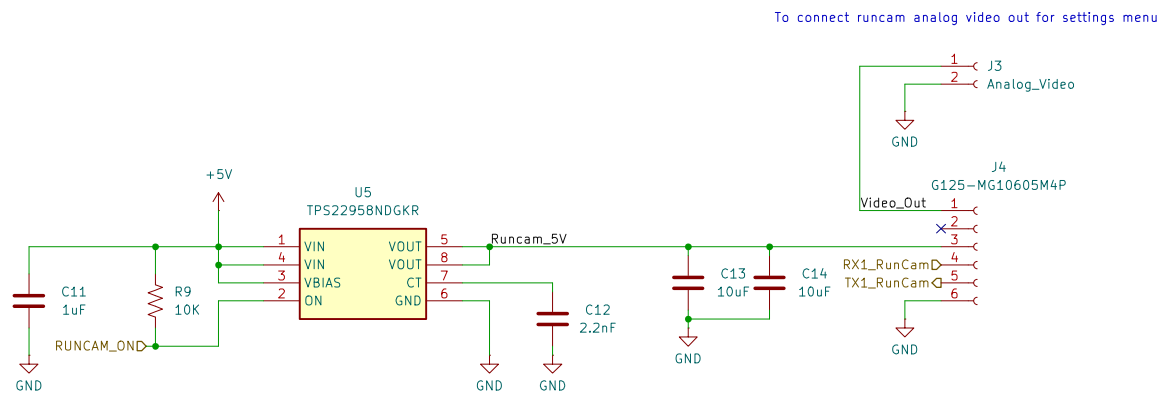
Size: A4

Date: 2026-06-08

Rev: B

KiCad E.D.A. 10.0.2

Id: 3/4



Waterloo Rocketry

Sheet: /RunCam/
File: runcam.kicad_sch

Title: Camera Interface Board

Size: A4

Date: 2026-06-08

Rev: B

KiCad E.D.A. 10.0.2

Id: 4/4